



Classifying the Variety of Customers' Online Engagement for Churn Prediction with a Mixed-Penalty Logistic Regression

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Abstract

Using big data to analyze consumer behavior can provide effective decision-making tools for preventing customer attrition (churn) in customer relationship management (CRM). Focusing on a CRM dataset with several different categories of factors that impact customer heterogeneity (i.e., usage of self-care service channels, service duration, and responsiveness to marketing actions), this research provides new predictive analytics of customer churn rate based on a machine learning method that enhances the classification of logistic regression by adding a mixed penalty term. The proposed penalized logistic regression prevents overfitting when dealing with big data and minimizes the loss function when balancing the cost from the median (absolute value) and mean (squared value) regularization. We show the analytical properties of the proposed method and its computational advantage in this research. In addition, we investigate the performance of the proposed method with a CRM dataset (that has a large number of features) under different settings by efficiently eliminating the disturbance of (1) least important features and (2) sensitivity from the minority (churn) class. Our empirical results confirm the expected performance of the proposed method in full compliance with the common classification criteria (i.e., accuracy, precision, and recall) for evaluating machine learning methods.

Keywords Big data · Business analytics · CRM · Machine learning · Penalized logistic regression

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1 Introduction

Due to the digitalization and availability of new technologies since the new millennium, online gambling has become an emerging trend in today's social entertainment activities, as shown by the industry's continuous growth. For example, the online poker market continues to grow globally, and its annual revenue is estimated to be approximately US\$ 3.6 billion (see Fiedler & Wilcke, 2011). More and more people use devices such as tablets, mobile phones, and personal computers to place bets remotely and participate in online gambling, with sports and virtual poker betting as the most popular types of activities. With the development of online gambling, the literature on the gambling industry, especially the online gambling industry, has gradually increased, and the research direction has become more diversified. Several studies on the gambling industry focus on the problem of gambling addiction (see Gainsbury et al., 2014; Johansen et al., 2019, for example), analysis of user psychology and behavior for website optimization (see Hing et al., 2015; Konietzny et al., 2018; Scott et al., 2019), and customer churn (Castro & Tsuzuki, 2015). After analyzing and comparing the traditional gambling industry and online gambling industry, Siemens and Kopp (2011) report that although consumers spend a lot of time gambling online, they do not spend too much money online. Due to rapid market updates and the intensification of industry competition, every online gaming platform is facing tremendous pressure from customers. Therefore, increasing numbers of operators are beginning to pay attention on how to effectively predict customer churn so as to take timely measures to retain customers who may churn and reduce platform losses. A large number of studies has considered customer churn across various fields and industries (see Coussement & De Bock, 2013; Milošević et al., 2017, for example).

Most markets have become saturated, and with the acceleration of product updates, market competition has become increasingly fierce (see Fiedler & Wilcke, 2011). Therefore, many companies are encountering a serious customer churn crisis. To solve this problem, they have shifted from a quote-centric strategy (sell as many products as possible) to a customer-centric retention method—that is, to retain as many customers as possible to reduce customer churn. Good churn management can effectively increase a company's revenue and help it establish a good reputation in the market (Amin et al., 2019). Customer churn prediction model based on a customer's historical data enables users to infer future customer churn probability. The model can also be used to identify customers whom the company wants to retain and provide targeted services for them. The model's analysis can also help the company find the main reasons that affect customer churn and solve the problem in a targeted manner (De Caigny et al., 2018).

The problem of customer churn is essentially a classification one. Understanding customer churn first requires understanding the business knowledge. According to the theory of CRM, churn is divided into voluntary churn, and non-voluntary churn (see Amin et al., 2019, and references therein). Voluntary churn is usually caused by products, companies, and competitors that make customers move to another product.

An example of non-volunteer churn is the relocation of a company or website. This study of customer churn will focus only on voluntary churn and to help determine the commonly used indicators of customer churn, we find relevant indicators so as to dig out insights using customer data. In this study we take the data of 7467 customers to build our classification model.

We propose a machine learning-based optimization method (namely, mixed penalty logistic regression model) to solve the predictive classification problem. Logistic regression is a common way to deal with binary classification. It is a kind of linear classifier and is widely used in various fields (see Affes & Hentati-Kaffel, 2019; Pesantez-Narvaez et al., 2020; Zhang & Trubey, 2019, for example). Even though it is apparent that a large number of different, more advanced, non-linear models and machine learning algorithms have been used for classification problems, logistic regression still represents the industry standard due to mandatory requirements about model explainability for practical purposes and a lack of understanding within artificial neural networks or black boxes of deep learning algorithms. This classification method can be generalized and also has high interpretability. In practice, as dimensionality increases, the appearance of the curse of dimensionality will cause the model to overfit. In order to prevent overfitting, a penalty term (l_1 or l_2 norm) is usually used. We propose a method to find an optimal mixing combination of l_1 or l_2 norms that is more accurate and suitable according to the loss function.

The results from our study show that when the targeted customers account for a relatively small proportion of the overall sample, logistic regression models without a penalty term will produce overfitting. Therefore, the findings of the proposed machine learning-based algorithm are twofold: first, for any given penalty value in the loss function, combined with the chosen mixture of l_1 and l_2 norms, we maximize the log-likelihood function; second, for the performance measure of interest, among all the combinations of the penalty coefficient and mixture norms maximizing the likelihood function, we choose the optimal one: a combination of the penalty and mixing coefficients that maximize the corresponding performance measure. Moreover, for such an optimal penalty coefficient and mixing combination of norms, we identify the most important features that maximize targeted performance. The robustness of the results is shown through a simulation study over four different methods using bootstrapped samples that differ regarding the choice of the training set versus the validating set in the proposed algorithm. When the enlarged training set is used to keep the same percentage of churners as in the original dataset, the algorithm is capable of significantly increasing the considered performance measure even with respect to a standard mild penalty term ($\lambda = 1$), independent of the mixing norm coefficient. Moreover, feature importance characteristics differ with respect to the chosen performance measure and the penalty term.

The results indicate that the importance of the main features is preserved for the overall methods used and for both the performance measures. However, the importance of other features changes depending on the method used, penalty term, and mixing norms coefficient. If the no-penalty model is used, then the most important feature from the proposed mixture penalized model is not detected in any of the methods used, independent of the performance measure of the interest. More precisely, the proposed algorithm manages to find an optimal combination of the

penalty function's parameter values for which the performance measure of interest is maximized. Consequently, the proposed method enables the company to emphasize the main reasons that affect customer churn, making it possible to solve the problem in a targeted manner.

We organize rest of this article as follows. Section 2 introduces the proposed mixed penalty logistic regression model and its estimation. Section 3 presents an empirical study with a customer's data to show how we conduct churn prediction with the proposed machine learning-based algorithm. We show that our proposed method indeed suffices for the goal of predictive churn classification in search of manageable tools for CRM. Section 4 summarizes and highlights future work.

2 Mix of Penalized Logistic Regression and Likelihood-Based Machine Learning Algorithm

For the classification of a binary variable Y (e.g., churn and retention), the generalized linear models with regularization can be applied with a coordinate descent method that is well defined for logistic, multinomial, and Poisson regressions (see Friedman et al., 2007; Wu et al., 2009, for example).

In this study, we start our analysis by considering the following optimization problem:

$$\min_{\Theta} \{L(\mathbf{X}, \mathbf{y}, \Theta)\}, \quad (1)$$

where $L(\mathbf{X}, \mathbf{y}, \Theta)$ is a loss function quantifying the statistical model parameterized by Θ with data $(\mathbf{X}, \mathbf{y})$. Our study aims to improve prediction accuracy and provide a performance measure of interest by shrinking the values of the regression coefficients or setting some coefficients to zero. By resembling the regression with regularization problem from Hastie et al. (2009) and measuring accuracy in terms of the mean-squared error, shrinking the values of the regression coefficients or setting some coefficients to zero introduces some bias, but reduces the variance of the predicted values and hence may improve the overall prediction accuracy (see Varma & Simon, 2006, for example).

The problem in Eq. (1) is a complex one that grows with the cardinality of the variables. Consequently, the possibility of finding a solution is not guaranteed. Moreover, the problem becomes indeterminate, because the number of features is larger than the number of past observations. In this age of big data, the number of variables representing the features that measure on a person or object increases rapidly and possibly becomes larger than the number of observations. The sparsity assumption (that is, a sparse statistical model) enables us to obtain an easier estimation and interpretation of the model in comparison to a dense or full model, which would help us tackle such problems and, more importantly, extract useful information and reproducible patterns from big datasets (Hastie et al., 2015). In this way, we aim to identify those features that play the most important role in exhibiting the strongest effect. The sparsity assumption greatly simplifies the computation, allowing for scalable algorithms with even millions of parameters. Consequently, we

analyze statistical methods that exploit sparsity to help recover a possible underlying pattern in a set of data for classification problems.

To this end, we consider the following optimization problem:

$$\min_{\Theta} \{L(\mathbf{X}, \mathbf{y}, \Theta) + \lambda \varrho(\Theta)\}, \quad (2)$$

where $\varrho(\Theta)$ is a penalty function on the parameter Θ added to the loss function of the optimization problem in Eq. (1), and its impact is controlled by a non-negative tuning parameter λ that minimizes the loss function L and penalizes the magnitude of the penalty function ϱ . On the one hand, when λ is equal to 0, there is no penalty for the coefficients of minor importance, and the result is equivalent to the classic model without dimensional restrictions. On the other hand, when λ grows, the influence of the dimensional restrictions gains weight and dominates Eq. (2). From the complex analysis theory, the necessary and sufficient conditions for a solution to Eq. (2) for specific loss and penalty functions can be derived. The system of equations to be solved in order to estimate the parameter Θ forms the Karush–Kuhn–Tucker (KKT) conditions for Eq. (2).

We thus consider the problem:

$$\min_{\Theta} \left\{ L(\Theta; \mathbf{y}, \mathbf{X}) + \lambda \varrho(\Theta) \right\}, \quad (3)$$

where we assume, for each $\mathbf{y} \in \mathbb{R}^N$, that the function $L(\cdot; \mathbf{y})$ is essentially smooth and essentially strictly convex on $\mathbb{R}^N$, or of Legendre type; see Rockafellar (1970).

In order to consider generalized linear models with a general penalty we take the loss function $L(\cdot; \mathbf{y})$ to be the negative log-likelihood function from a generalized linear model for $\mathbf{X}$. Just as in Taylor and Tibshirani (2011) and Ali and Tibshirani (2019), we first consider the generalized lasso problem as follows:

$$\min_{\Theta} \left\{ -\frac{1}{N} \mathcal{L}(\Theta; \mathbf{y}, \mathbf{X}) + \lambda \|\mathbf{D} \Theta\|_1 \right\}, \quad (4)$$

where $\mathbf{y} \in \mathbb{R}^N$ is the vector of outcomes with N samples, $\Theta = (\theta_0, \Theta) \in \mathbb{R} \times \mathbb{R}^M$ are the population parameters, $\mathbf{X} = [\mathbf{x}_i] \in \mathbb{R}^{N \times M}$ is the matrix of predictors with M features, $\mathbf{D} \in \mathbb{R}^{M \times M}$ is a penalty matrix, $\lambda \geq 0$ is a tuning parameter, and the specific form of the log-likelihood $\mathcal{L}$ varies according to the generalized linear model. The generalized lasso of Eq. (4) encompasses several well-studied problems as special cases corresponding to different choices of $\mathbf{D}$ with the following properties.

Property 1 If $\lambda = 0$, then we retrieve the generalized linear model without regularization or penalty. The solution to Eq. (4) thus exists, and every solution $\hat{\Theta}$ gives rise to the same fitted value $\mathbf{X}\hat{\Theta}$.

Property 2 Consider the following log-likelihood function:

$$\mathcal{L}(\Theta; \mathbf{y}, \mathbf{X}) = \sum_{i=1}^N [y_i(\theta_0 + \mathbf{x}_i^T \Theta) - \log(1 + e^{\theta_0 + \mathbf{x}_i^T \Theta})], \quad (5)$$

where $\Theta = (\theta_0, \Theta) \in \mathbb{R} \times \mathbb{R}^p$ are the population parameters, and $\mathbf{x}_i^T$ is the transpose vector of the i -th row of the predictor matrix $\mathbf{X}$, $i = 1, \dots, N$. Let the penalty function $\varrho(\Theta)$ be the l_1 regularization:

$$\varrho(\Theta) = \|\mathbf{D}\Theta\|_1. \quad (6)$$

If the general loss function in Eq. (4) is the negative log-likelihood given in Eq. (5) and $\mathbf{D} = \mathbf{I} \in \mathbb{R}^{M \times M}$, in Eq. (6), then it is a lasso-penalized logistic regression that always has optimal solutions (see Ali & Tibshirani, 2019).

Therefore, based on Properties 1 and 2 we derive the following theorem.

Theorem 1 Suppose the general loss function $L(\cdot)$ is a hinge loss such that:

$$L(\mathbf{X}, \mathbf{y}, \Theta) = \sum_{i=1}^N [1 - y_i(\theta_0 + \mathbf{x}_i^T \Theta)]_+ \quad (7)$$

and $\mathbf{D} = \mathbf{I} \in \mathbb{R}^{M \times M}$ in Eq. (6). The optimization problem in Eq. (2) then leads to l_1 sparse solutions.

Proof The hinge loss function defined in Eq. (7) is essentially smooth and essentially convex. Denote with $\text{null}(\mathbf{A})$ the null space of a matrix $\mathbf{A}$. In the case when $\mathbf{D} = \mathbf{I} \in \mathbb{R}^{M \times M}$, it follows that:

$$\text{null}(\mathbf{D}) \subseteq \text{null}(\mathbf{X}). \quad (8)$$

According to Lemma 16 in Ali and Tibshirani (2019), the condition in Eq. (8) is sufficient for an existing solution to problem (3). Consequently, an estimated $\hat{\Theta}$ suffices the solution to problem (3) fitted by $\mathbf{X}$. $\square$

For our analysis, we shall consider the logistic regression with elastic net penalty as a generalization of the lasso penalty by adjusting the loss function. The elastic net combines a squared l_2 penalty with the l_1 penalty in order to overcome the problem of possibly highly correlated features. Following the approach in Zou and Hastie (2005), we consider the elastic net as a compromise between the ridge and lasso penalties by solving the following minimization problem:

$$\min_{\theta_0, \Theta} \left\{ -\frac{1}{N} \sum_{i=1}^N [y_i(\theta_0 + \mathbf{x}_i^T \Theta) - \log(1 + e^{\theta_0 + \mathbf{x}_i^T \Theta})] + \lambda \left[\frac{1}{2} (1 - \alpha) \|\Theta\|_2^2 + \alpha \|\Theta\|_1 \right] \right\}, \quad (9)$$

where $y \in \mathbb{R}^N$ is a binary response vector, $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_i)$ is a feature matrix, $\theta_0 \in \mathbb{R}$ is an intercept term, $\Theta \in \mathbb{R}^M$ is a vector of regression coefficients, $\mathbf{x}_i^T$ is the transpose vector, $\alpha \in [0, 1]$, and λ is the regularization weight. Considering sparsity

enables us to solve the optimal problem in Eq. (9) with different perspectives from complicated procedures when searching for the solution of Eq. (1).

Remark 1 When $\alpha = 1$, the penalty applied in Eq. (9) reduces to the l_1 norm or lasso penalty, and when $\alpha = 0$, it reduces to the squared l_2 norm corresponding to the ridge penalty.

In our analysis, we code the dichotomous binary variable Y as a 0/1 response and assume that the binary variable Y is defined on some probability space $(\Omega, \mathcal{F}, P)$.

2.1 From Maximum-Likelihood Inference to Machine Learning Inference

This subsection proposes a machine learning-based algorithm to deal with the sparse logistic regression optimization for Eq. (9) jointly with finding the corresponding feature importance.

For estimation purposes we consider a statistical experiment $(\Omega, \mathcal{F}, \{P_{\Theta} : \Theta \in \Theta\})$ for the parameter space $\Theta = \{\Theta \in \mathbb{R}^M\}$. The data are generated under the true probability measure P_{Θ_0} with some $\Theta_0 \in \Theta$. The expectation with respect to P_{Θ} is denoted by $E_{\Theta}[\cdot]$ and with respect to P_{Θ_0} simply by $E[\cdot]$. We assume there is a process Y in which the conditional distribution given X is multinomial under P_{Θ_0} . In order to find an estimator $\hat{\Theta}_0$ from $\mathbf{x}_1, \dots, \mathbf{x}_N$, we consider the log-likelihood in Eq. (5) for N observations as follows:

$$\mathcal{L}(\Theta; \mathbf{y}, \mathbf{X}) = \sum_{i=1}^N \log p_{y_i}(\mathbf{x}_i; \Theta), \quad (10)$$

where

$$p_{y_i}(\mathbf{x}_i; \Theta) = P(Y_i = k | \mathbf{X} = \mathbf{x}_i; \Theta), \quad k = 0, 1. \quad (11)$$

Using the 0/1 code for the response y_i , the log-likelihood in Eq. (10) can be written as:

$$\begin{aligned} \mathcal{L}(\Theta; \mathbf{y}, \mathbf{X}) &= \sum_{i=1}^N \left\{ y_i \log p(\mathbf{x}_i; \Theta) + (1 - y_i) \log(1 - p(\mathbf{x}_i; \Theta)) \right\}, \\ &= \sum_{i=1}^N \left\{ y_i (\theta_0 + \mathbf{x}_i^T \Theta) - \log(1 + e^{\theta_0 + \mathbf{x}_i^T \Theta}) \right\}, \end{aligned} \quad (12)$$

where $\Theta = (\theta_0, \Theta) \in \mathbb{R} \times \mathbb{R}^M$. The exponential value of regression coefficients, e^{Θ} , gives the odds ratio, and reflects the impact of the risk factor of churn (i.e., $Y = 1$). For maximizing the log-likelihood in Eq. (12), a variety of numerical optimization algorithms, from a simple gradient descent to more complicated second-order methods of EM algorithms, can be used to determine the optimal parameter values; see Koslovsky et al. (2018).

An alternate approach for computing the solution to the elastic net problem in Eq. (9) is a coordinate descent, which is designed to fit regularized models. A coordinate descent is an iterative algorithm in which the update from θ^l to θ^{l+1} is

done choosing a single coordinate to update and then performing a univariate minimization over this coordinate. One may consider a function $f : \mathbb{R}^M \rightarrow \mathbb{R}$ that is differentiable and strictly convex in each coordinate. If coordinate k is chosen at iteration l , then the update is given by:

$$\theta_k^{l+1} = \arg \min_{\theta_k} f(\theta_1^l, \theta_2^l, \dots, \theta_{k-1}^l, \theta_k, \theta_{k+1}^l, \dots, \theta_M^l), \quad (13)$$

and $\theta_j^{l+1} = \theta_j^l$ for $j \neq k$. Furthermore, a typical choice would be to cycle through the coordinates in some fixed order.

The elastic net in Eq. (9) is convex in the pair $(\theta_0, \Theta) \in \mathbb{R} \times \mathbb{R}^M$, and thus the coordinate descent is particularly effective. In Eq. (9) we include an unpenalized intercept, and the optimal intercept is calculated centering the covariates $\mathbf{x}_{ij}$ and then calculating:

$$\hat{\theta}_0 = \bar{y} = \frac{1}{N} \sum_{j=1}^N y_j. \quad (14)$$

Equation (14) specifies that:

$$\sum_{j=1}^N y_j = \sum_{j=1}^N p(\mathbf{x}_j; \Theta), \quad (15)$$

and so the expected number of class 1 matches the observed number and hence also for class 0. Having solved for $\hat{\theta}_0$, it remains to compute the optimal model parameter vector $\hat{\Theta} = (\hat{\theta}_1, \dots, \hat{\theta}_M)$.

To enable the procedure in Eq. (13) to converge to the global minimum of a convex function, along with the use of statistical regularizers, the continuous function f does not need to be differentiable. In this case, special attention is required when using coordinate minimization. For a problem structure that ensures good behavior of coordinate minimization in Eq. (13), we consider the following loss function:

$$f(\theta_1, \dots, \theta_M) = -\frac{1}{2N} \sum_{i=1}^N \log p_{y_i}(\mathbf{x}_i; \theta_1, \dots, \theta_M) + \lambda \sum_{j=1}^M \left[\frac{1}{2} (1 - \alpha) \theta_j^2 + \alpha |\theta_j| \right], \quad \alpha \in (0, 1). \quad (16)$$

For any $\alpha \in (0, 1)$ and $\lambda > 0$, the loss function f defined by Eq. (16) is strictly convex, and so a unique solution exists irrespective of the correlations or duplications in the features X_j , $j = 1, \dots, M$. In order to incorporate the lasso penalty or l_1 -norm in our general optimization problem, we consider the additive decomposition property of f given by Eq. (16) with the first component being a differentiable and convex multivariate function of $\Theta \in \mathbb{R}^M$ and the second component a convex, but not necessarily differentiable univariate function.

Denoting f_i with the i -th component of the loss function $\mathcal{L}(\cdot)$ from Eq. (10), the first summand in Eq. (16) is then:

$$f_i(\cdot; \Theta) = \log p_{y_i}(\cdot; \Theta). \quad (17)$$

For estimating the parameters of the elastic net problem given by Eq. (9) we consider the composite case where an additional regularization function $h : \mathbb{R}^M \rightarrow \mathbb{R}$ is added to the loss function:

$$h(\Theta) = \sum_{j=1}^M h_\alpha(\theta_j) = \sum_{j=1}^M \left[\frac{1}{2} (1 - \alpha) \theta_j^2 + \alpha |\theta_j| \right], \quad \forall \alpha \in (0, 1). \quad (18)$$

The key property of the additive decomposition in Eq. (16) is the separability of the second component—namely, the non-differentiable component of h given by:

$$h(\theta_1, \dots, \theta_M) = \sum_{j=1}^M h_\alpha(\theta_j), \quad (19)$$

where $h_\alpha : \mathbb{R} \rightarrow \mathbb{R}$ represents the penalty applied to an individual coefficient that is given by:

$$h_\alpha(\theta_j) = \frac{1}{2} (1 - \alpha) \theta_j^2 + \alpha |\theta_j|, \quad \forall \alpha \in (0, 1). \quad (20)$$

As shown in Tseng (1998, 2001), for any convex loss function f with the separable structure given by Eq. (16), the coordinate descent algorithm in Eq. (13) is guaranteed to converge to the global minimizer. This result implies that the coordinate descent is a suitable algorithm for the whole family of elastic net problems given by Eq. (9). Moreover, in cases where the penalty h in Eq. (18) is convex and potentially non-differentiable, and where the proximal operation of h is easy to compute, few incremental gradient methods are applicable; see Defazio et al. (2014).

2.2 Machine Learning-Based Sparse Likelihood Algorithm

In order to solve the penalized log-likelihood problem in Eq. (9), the coordinate descent is not applied directly as in Eq. (13), because the optimizing values along each coordinate are not explicitly available and require a line search. The function $h : \mathbb{R}^M \rightarrow \mathbb{R}$ given by Eq. (18) is convex but potentially non-differentiable, and therefore, we propose a new machine learning-based sparse likelihood algorithm based on the SAGA algorithm (Defazio et al., 2014) as an incremental gradient method to support non-strongly convex composite objects. As emphasized in Defazio et al. (2014), although any incremental gradient method can be applied to non-strongly convex problems through the addition of a small quadratic regularization, the amount of regularization is an additional tunable parameter that the SAGA algorithm avoids. More important, when a specific step size is used in the incremental gradient method, the SAGA algorithm automatically adapts to the level of strong convexity that is naturally present.

The idea of the proposed method in Algorithm 1 stems from the recent advances in minimization of strongly convex finite sums. Thus, algorithms based on machine learning arise spontaneously in the empirical risk minimization framework when a

common finite sum structure is present. We summarize the proposed method in Algorithm 1 that is more suitable than the conventional stochastic average gradient descent methods by depriving it from the presence of a convex, but potentially non-differentiable composite or proximal term in the elastic net problem (9) by denoting an additional regularisation function.

Algorithm 1 Machine learning-based sparse likelihood algorithm

Input: The dataset, including the training $\mathcal{T}$, validation $\mathcal{V}$ and testing datasets, $\mathcal{S}$; set of mixture coefficients $\alpha \in [0, 1]$; set of penalty coefficients $\lambda \in [0, 1]$.

Output: classification performance; i.e., accuracy and recall.

```

1: for  $k \in \{A, R\}$  do
2:   for  $\alpha = 0$  to 1 do
3:     for  $\lambda = 0$  to 1 do
4:       Use  $\mathcal{T}$  to solve  $\Theta_{k,\alpha,\lambda} = \arg \min_{\theta_0, \theta} \left\{ -\frac{1}{|\mathcal{T}|} \sum_{\mathbf{x}_i, y_i \in \mathcal{T}} f(\mathbf{x}_i) + \lambda h_\alpha(\theta) \right\}$  by Algo. 2.
5:       Use  $\mathcal{V}$  to compute  $a_{k,\alpha,\lambda} = \mathcal{A}_k(\Theta_{k,\alpha,\lambda}, \mathcal{V})$ 
6:     end for
7:      $\lambda_{k,\alpha}^* = \arg \max_\lambda a_{k,\alpha,\lambda}$ 
8:   end for
9:   Use  $\mathcal{S}$ , and  $\Theta_{k,\alpha,\lambda_{k,\alpha}^*}$  for  $\alpha \in [0, 1]$ , to evaluate the accuracy and recall  $\mathcal{A}_k(\Theta_{k,\alpha,\lambda_{k,\alpha}^*}, \mathcal{S})$ .
10: end for
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The implementation of the SAGA method for dense gradients in our machine learning-based algorithm is well understood and directly implementable. However, for sparse problems with a quadratic regularizer, the just-in-time updating of the elements of the iterate vector can be a little bit involved, but it is a necessary step aiming to avoid full dense vector operations at each iteration. Consequently, following the Defazio et al. (2014) approach, instead of scaling Θ at each step, a separate scaling constant can be used. Therefore, the proposed algorithm (in Algorithm 1) efficiently implements the SAGA incremental gradient method for the convex combination of the lasso and ridge penalty in the sparse logistic regression with elastic net penalty in Eq. (9).

Given a training set $\mathcal{T} = \{\mathbf{x}_i, y_i\}$, Θ can be derived in the training procedure. More precisely, we start with some known initial vector $\Theta^0 \in \mathbb{R}^M$ and known derivatives $f'_{\Theta^0}(\mathbf{x}_i)$ for each $i = 1, \dots, |\mathcal{T}|$, where $f(\cdot)$ is strongly convex (specifically, in the elastic net penalty regression problem (9) it is the logistic function) with constant μ and has Lipschitz continuous derivatives with constant L . We use an auxiliary table $\mathbf{a}$ to store derivatives for each training sample, where $\mathbf{a} \in \mathcal{R}^{|\mathcal{T}| \times M}$. During the training process, the values in $\mathbf{a}$ are updated when convergence improves. Following Defazio et al. (2014) we consider the binary classification problem with only a single floating point value to be stored instead of a full gradient vector. The estimate updates in the SAGA method starting from $k = 0$ with a specific step size γ to be done in the following way.

- (1) Randomly choose j from the uniformly distributed gradient sample, $j = 1, \dots, |\mathcal{T}|$;
- (2) Compute $f'_{\Theta^k}(\mathbf{x}_j)$ and store it in $\mathbf{a}$ after updating, while keeping all other entries $i \neq j$ in $\mathbf{a}$ unchanged;

(3) Compute w^{k+1} with $f'_{\Theta^k}(\mathbf{x}_j)$, $f'_{\Theta^{k-1}}(\mathbf{x}_j)$, and $\mathbf{a}$ as follows:

$$w^{k+1} = \Theta^k - \gamma \left[f'_{\Theta^k}(\mathbf{x}_j) - f'_{\Theta^{k-1}}(\mathbf{x}_j) + \frac{1}{|\mathcal{T}|} \sum_{i=1}^{|\mathcal{T}|} \mathbf{a}[i] \right], \quad \gamma = \frac{1}{2(L + \mu|\mathcal{T}|)}; \quad (21)$$

(4) Use w^{k+1} to update Θ^{k+1} , such that:

$$\Theta^{k+1} = \text{prox}_{\gamma}^h(w^{k+1}) = \arg \min_{\Theta} \{h_{\alpha}(\Theta) + \frac{1}{2\gamma} \|\Theta - w^{k+1}\|^2\}. \quad (22)$$

The critical step of the proposed algorithm is to split the data into training set $\mathcal{T}$, validation set $\mathcal{V}$, and test sets $\mathcal{S}$ when determining the optimal penalty coefficient λ^* via cross-validation on the validation set. In this way the accuracy or any performance measure of interest can be computed on the test sets, and the corresponding feature importance can be obtained based on the whole dataset with the optimal λ^* parameter.

The parameters, or more specifically the possible value of α in the elastic net penalty in Eq. (9) and parameter λ , characterize the slackness of the whole process. Different values of λ from a grid of values are tested to find the optimum λ^* that minimizes the negative log-likelihood function $\mathcal{L}(\cdot)$ in Eq. (5) and penalizes surplus and possibly correlated features.

In order to choose the right evaluation measure for the proposed algorithm, we account for the fact that the typically used performance measure for classification problems—accuracy, being the ratio of correctly classified items—is generally not the preferred performance measure for classifiers, especially when dealing with skewed and unbalanced datasets. From a business viewpoint, the accuracy performance measure is not appropriate either. Namely, there is a significantly different potential financial loss by incorrectly classifying the non-churners from churners. More specifically, when incorrectly classifying non-churners ($y_i = 0$) as churners ($y_i = 1$), although they are actually non-churners, the only financial loss that the company may have is the cost of self-promoting in order to shape customers' decision-making and reactivate their service for staying with the company. However, when wrongly classifying churners ($y_i = 1$) as non-churners ($y_i = 0$), while they are actually churners, the potential financial loss for the company is much higher. The company has a faulty belief that the customer has been engaged with the company as a paid customer and has no strict motivation to influence the client to remain with the company.

It is consequently not equally profitable to measure correctly classified churners and non-churners either in the same way or at the same cost. We need to consider a performance measure that is not going to reject many churners while at the same time allowing a smaller kind of error when a few non-churners show up as churners. The company in question needs to rely on maximizing a performance measure that assures a smaller number of false alarms (when categorizing a non-churner as a

churner), whereas most real churners are getting detected. In that respect, we propose considering the recall performance metric, defined as:

$$R = \frac{TP}{TP + FN}, \quad (23)$$

where TP is the number of true churners (positive) classified and FN is the number of false non-churners (negative) classified. Representing the ratio of positive instances ($y_i = 1$) that are correctly detected by the classifier, the recall performance measure represents the sensitivity or true positive rate.

The steps of the proposed algorithm are described in detail in Algorithm 2. The outputs of the algorithm are the optimal regularization weight for a specific varying parameter α in the elastic net, denoted by λ^* , the classification performance in terms of accuracy and recall for that optimal value and estimated features' importance based on the corresponding performance measures. The requirements to apply this algorithm are: (1) the dataset splits (training and testing), (2) set of discrete values of the regularization weight λ from the interval $(0, \infty)$, (3) a set of values of the percentage of l_1 norm present in the elastic net α from 0% to 100%, and (4) the performance measure function. The algorithm solves the optimization problem on the training subset for each of the different values of λ and α . Once the optimization problem is solved, we get a set of performance measure values on the test set, which are as many as the values for λ . Subsequently, we make the selection of the optimal λ^* as the one that maximizes the considered performance measure. Due to this selection, it is possible to obtain the best penalty in terms of the performance measure. Once the best λ^* is obtained, we apply the algorithm on the test set and derive the features' importance based on the accuracy and recall performance measures. Feature importance analysis can be conducted directly following Michel et al. (2011) and Korobov (2020) to determine a feature's relative importance.

In our study, we apply the concept of permutation importance. Each feature is permuted — that is, the values for that feature are shuffled randomly until the specified feature no longer provides useful information to the model. For managerial considerations, we interpret feature importance as a measure on how much the performance score of interest decreases when a corresponding feature is not available. Consequently, negative values may also arise, emphasizing a score increase when the feature is permuted. This happens, because the feature permutation actually improves the score. In this way we can evaluate feature importance not only based on its performance scoring, but also for its contribution to the decision process.

Algorithm 2 SAGA-Proximal gradient algorithm

Input: Training dataset $\mathcal{T} = \{(\mathbf{x}_i, y_i)\}, 1 \leq i \leq |\mathcal{T}|$; the step size γ ; loss function $f(\cdot)$, see Equation (12); regularization function $h_\alpha(\cdot)$, see Equation (18).

Output: Θ

```

1:  $k = 0$ 
2: Initialize  $\Theta^0$ .
3: Initialize an auxiliary table  $\mathbf{a}$  for derivatives.
4: for  $i = 1$  to  $|\mathcal{T}|$  do
5:   Compute the derivative  $f'_{\Theta^0}(\mathbf{x}_i)$ 
6:   Update  $\mathbf{a}[i] \leftarrow f'_{\Theta^0}(\mathbf{x}_i)$ .
7: end for
8: while  $\Theta^k$  not converged do
9:   Randomly sample  $j$  from  $\{1, \dots, |\mathcal{T}|\}$ .
10:  Regarding to  $\Theta^k$ , compute the derivative  $f'_{\Theta^k}(\mathbf{x}_j)$ .
11:  Update  $\mathbf{a}[j] \leftarrow f'_{\Theta^k}(\mathbf{x}_j)$ 
12:  Compute  $w^{k+1} = \Theta^k - \gamma \left[ f'_{\Theta^k}(\mathbf{x}_j) - f'_{\Theta^{k-1}}(\mathbf{x}_j) + \frac{1}{|\mathcal{T}|} \sum_{i=1}^{|\mathcal{T}|} \mathbf{a}[i] \right]$ .
13:  Compute  $\Theta^{k+1} = \text{prox}_{\gamma}^h(w^{k+1}) = \arg \min \{h_\alpha(\Theta) + \frac{1}{2\gamma} \|\Theta - w^{k+1}\|^2\}$ .
14:    $k+ = 1$ 
15: end while

```

3 Empirical Study

3.1 The Data

In order to illustrate the empirical features of the proposed algorithm (see Algorithm 1), we use a proprietary dataset provided by Hattrick Sports Group's internal databases.¹ The proposed algorithm estimates the sparse logistic regression model coefficients Θ through optimization of two different performance measures (accuracy and recall). Moreover, by using both accuracy and recall as performance measures, feature importance is produced based on the estimated coefficients. The predictive ability of the proposed algorithm jointly with the feature importance varies significantly from the non-regularized logistic regression case, but they are similar along the mixing parameter α in the elastic net specification. The robustness of the results is confirmed in the simulation exercise in Sect. 3.3.

Hattrick Sports Group supports gaming operations across Europe, providing both business-to-business (B2B) and business-to-consumer (B2C) services. Data for the analysis herein are taken from one of the markets in which the company operates. For future reference, we name the company XComp. Data fields were masked in order to protect the classified data, but they still remained usable for the purpose of the analysis.

Since June 2015 computer betting operators have offered the possibility of online betting and gaming services. More specifically, betting operators offer the opportunity to bet on the results of sports matches (football, basketball, cricket, handball, table tennis, etc.) and also offer the opportunity to bet on virtual games (for

¹ <http://www.hattricksg.com/>.

example, virtual dogs, horses, and bingo). Sports betting is additionally split into Sport Live, Sport Prematch, and Sport Mixed betting. Sport Mixed betting is when the player on the betting slip has bets that are both live and prematch. XComp enables customers to react to changing circumstances by making bets during the event.

In order to be available to as many users as possible, the company offers its service through four channels: desktop web, mobile web, Android, and iOS applications. During the registration process, each customer is assigned an account that will be used for further betting transactions (both inflow and outflow from the betting). The company offers several different channels through which users can deposit or withdraw their money from and to the XComp online account: staffed counters (over-the-counter transactions), credit cards, banks, and external payment providers.

The online betting industry provides a good example to reveal important patterns in data, because any online operator can easily record and access the relevant data about its customers and overall performance. The data used for this study consist of datasets about the online betting activities of betting operator XComp during from November 1, 2017, to March 1, 2018. The data contain entities, attributes, and facts as shown in Table 1. We observe the data of players' activity X in the specified period and group the data by a single player. For those determined players, we observe if they are active or not in later months in order to determine and/or form the dichotomous binary variable Y .

There is also an additional dataset for each dimension identified in Table 1, such as Customers (table consisting of customers' personal information, like gender, city, age, registration date, and others) or Products (provider information, product name, etc.). As of September 30, 2018, XComp had 48,466 registered customers. The majority of customers are male, representing 80% of registrations. In terms of age, the customers are more diversified, with more than 50% of customers aged between 18 and 45 years old. Descriptive statistical analysis shows 1,606 daily active customers (on average), with maximum value of 2,625 recorded from September 2018 onward with a coefficient of variation of 24%. Expected daily turnover equals \$97,000, with a standard deviation of \$35,000 and a skewness of 1.1, indicating a positively skewed distribution. Gross gaming revenue (GGR) ranges from \$19,000 to \$35,000, with an average daily GGR of \$8,130 and 76% of variability around the mean. With a higher standard deviation relative to the mean, \$6,630 distorts the reliability of the mean as a measure of central tendency for GGR.

Description of the variables used in the analysis is given in Table 2. The variable Ticket-Coverage reflects a customer's ability to place a ticket in the next 30 days without making a new wallet deposit. Balance amount itself does not represent the same for each customer in terms of potential activity. For instance, a wallet balance of \$10 for a customer with an average ticket size of \$5 leaves room for two additional tickets, but no ticket for a customer with an average ticket size of \$15. Descriptive statistics of the features and the churn response variable are given in Table 3.

The data analysis is split into three parts. In the first part, we consider the no-regularization logistic regression case—that is, we solve the problem (9) with $\lambda = 0$. The second part of the analysis considers problem (9) with a typically default

Table 1 Transactions

Column name	SQL server data type	Nullable	Description
TransID	Integer (INT)	No	Unique identifier of the transaction
TransactionTime	Date and Time (DATETIME2)	No	Time when the transaction occurred
TransactionType	Integer (INT)	No	Unique identifier of a transaction type (1—bet, 2—win, 3—lost, 4—deposit, 5—withdrawal)
TransactionOrigin	Unicode string data (NVARCHAR)	No	Differentiates betting transactions from wallet deposit transactions
TicketID	Integer (INT)	No	Unique identifier of ticket
CustomerID	Integer (INT)	No	Unique identifier of a customer
Product	Unicode string data (NVARCHAR)	Yes	Product on which customer placed a bet, Sports, Live, Virtual Games
Device	Integer (INT)	No	Device used when using a service (iOS app, Android App, Mobile Web, Desktop Web)
Amount	Decimal	No	Bet amount (on a ticket level)

regularization weight $\lambda = 1$. Finally, the third part of the analysis considers the problem (9) with a general regularization weight λ and applies the proposed algorithm from Sect. 2.2 aiming to find the optimal regularization weight λ^* that maximizes the performance measure in question.

For each of the chosen performance measures defined in (23), we use a set of discrete algorithmic parameter values for the regularization weight λ from the interval $[1, \dots, 10^{10}]$, where we take 20 equidistant points on the logarithmic scale on that interval. Moreover, in all three cases of the analyzed model (i.e., $\lambda = 0$, $\lambda = 1$, and finding λ^*), the corresponding performance measure is used to derive the features' importance. The optimization problems have been solved using the Python library scikit-learn, where optimization algorithm SAGA is already implemented as an optimizing option. All optimizations have been initialized with a specific random seed.

3.2 The Results

To help illustrate the algorithm's application, in Table 4 we first focus on the performance of the algorithm for an initially predefined regularization weight $\lambda = 1$ using both performance measures defined in (23). Alternative restrictions or enlargements of the full dataset used for training and testing are possible; for example, the original dataset used for training and a bootstrapped (with replacements) dataset that is ten times bigger in size with the same percentage of churners as in the original dataset for testing, or with a balanced percentage of churners. Such alternative possibilities for training and testing (Method 1–Method 4) lead to similar key features regarding performance measures throughout the choice of the elastic net varying parameter α in the elastic net problem (9).

Table 2 Description of the features and response variables

Variable	Description	Example
Age	The difference between birthday and selected date in years; it reflects how old is the customer at the time of running the model; in this case, 01.03.2018	Birthday: 02.02.1987 Selected date: 01.03.2018 Age: 31
Life-Time-In-Months	The difference between registration date and selected date (in this case, selected date was 1.3.2018, because we want to predict which customers will churn in the following 30 days; that is, from 01.03.2018 to 01.04.2018 according to the customer's behavior in the previous 4 months, from 01.11.2017 to 01.03.2018)	Registration date: 01.01.2017 Selected date: 01.03.2018 Life-Time-In-Months: 14
Life-Time-Active-Months	During the customer's lifetime (time from registration date onwards), the customer does not have to be active each month. The variable counts the number of months in which the customer has placed at least one bet. Most loyal customers return from month to month, and so for such a group of customers, variable Life-time-Active-Months = Life-Time-In-Months	Customer C, placed 6 sports tickets since the registration date 01.04.2017. He placed 2 single tickets (tickets with single bet) in July 2017, 3 tickets with 2 bets in November 2017, and 1 ticket with 3 bets in February 2018 Life-Time-Active-Months: 3
Activity	Measure that counts the number of active months in customer lifetime, but puts more weight on a recent history	Calculated as $\sum_{i=1}^N \frac{I_{\text{activity}}}{\log(9+x_i)^{4.5}}$, where x_i represents the date difference between the active month and the selected date (March 2018), measured in months. For instance, suppose we have the case of two customers, A and B, both registered in October 2017. If customer A was active in November 2017 and December 2017 ($x_i = 3$ and 4, respectively), and customer B in January and February 2018 ($x_i = 2$ and 1), then both customers were active for two months, but customer B has higher activity value (1.833) compared to customer A (1.325), because he was active in more recent history.
Days-From-Last-Deposit	The difference between the last deposit date (transfer of money to a customer's XComp online account) and selected date (in the case 01.03.2018), measured in days	Customer C placed three deposits since the registration date, 01.04.2017. He placed deposits (transfer of money to his Xcomp online account) on 03.04.2017 in amount of \$30, 05.11.2017 in amount of \$10, and 01.02.2018 in amount of \$20; Days-From-Last-Deposit: 28
Avg-Deposit	The average amount of customer's transfers to XComp betting account	For customer C, Avg-deposit is \$20
Life-Time-Count-Deposits	The number of transfers to XComp online betting account	For customer C, Life-Time-Count-Deposits: 3
Life-Time-Deposit	The total deposited amount to XComp betting account from registration date onwards	For customer C, Life-Time-Deposit = \$60

Table 2 continued

Variable	Description	Example
Sport-Tickets	The number of sport tickets placed in previous 4 months (in the case, from 01.11.2017 to 01.03.2018)	For customer C, Sport-Tickets: 4
Mobile-Ratio	The share of wager amount placed over the mobile device, out of total wager amount (mobile wager + desktop web wager)	Wager – desktop web: \$35, Wager – mobile: \$15, Total Wager: \$50 Mobile-Ratio: 0.3
Live-Betting-Ratio	The share of wager amount on sports betting placed on in-play (live) matches, out of total wager amount placed on sports betting	Wager – sport prematch: \$10, Wager – sport mixed tickets: \$5, Wager – sport live tickets: \$35, Total Wager Sport: \$50 Live-Betting-Ratio: 0.7
Days-Active	Number of days when customer placed at least one bet in the observed period of time (01.11.2017–01.03.2018)	For customer C, Days-Active: 2
Last-Period-Activity	Ratio of total active days in observed period of time and number of active days in last 30 days	For customer C, Last-Period-Activity = 0.5.
Bonus-Cnt	Number of times when customer received a bonus amount	
Bonus-Amount	Total bonus amount received in the observed period, from 01.11.2017 to 01.03.2018	
New-Player	If the customer is registered in previous 30 days from selected date (01.03.2018), then 1, else 0	For customer C, New-Player = 0
Avg-Ticket-Size	Average ticket size, total wager amount divided by the number of tickets placed	For customer C, Avg-Ticket-Size = \$8.33
Avg-Bet-Count	Average number of bets placed on a sport ticket, rounded	For customer C, Avg-Bet-Count = 2
Distinct-Products	Number of different XComp services purchased by a customer. Xcomp offers two main betting services, betting on sports events and betting on virtual games	For customer C, Distinct-Products: 1 (customer placed bet only on sport).
Balance	Spendable amount on 01.03.2018	Wallet balance = Winnings – wager + deposits – withdrawals + bonus amount
Ticket-Coverage	Ratio of customer's wallet balance and customer's average bet size	If the customer has a balance of \$20 and plays \$5 ticket on average, Ticket Coverage = 4
Churn	Churn label, 0 if the player placed at least one bet from 01.03.2018 to 01.04.2018, else 1	

The key features of the optimal accuracy and recall performances, output by the proposed algorithm (for $\lambda = 1$) vary across the percentage of l_1 (correspondingly l_2) norm. The variations can be substantial, but they also reveal some consistencies dependent on the choice of the percentage of the original dataset used for training and testing. The percentages of l_1 norm for which Table 4 reports optimal performances in terms of accuracy and recall are $\{0\%, 10\%, \dots, 90\%, 100\%\}$. For all the

Table 3 Descriptive statistics of features and the churn response variable

	Mean	Std	Min	Q-25%	Q-50%	Q-75%	Max
Age	31.438	10.384	18.000	24.000	29.000	36.000	94.000
Life-Time-In-Months	22.805	16.658	1.000	6.000	18.000	44.000	45.000
Life-Time-Active-Months	12.061	11.253	0.000	3.000	8.000	18.000	45.000
Activity	4.444	2.874	0.000	1.866	3.855	6.641	11.000
Days-From-Last-Deposit	32.423	67.744	1.000	3.000	11.000	44.000	1352.000
Avg-Deposit	18.013	59.485	0.000	3.515	6.341	13.000	2250.000
Life-Time-Count-Deposits	114.530	221.831	0.000	8.000	31.000	115.000	3373.000
Life-Time-Deposit	1774.747	5808.145	0.000	43.675	218.750	1113.860	130562.460
Sport-Tickets	230.059	514.507	0.000	15.000	59.000	215.000	10087.000
Mobile-Ratio	0.761	0.375	0.000	0.600	1.000	1.000	1.000
Live-Betting-Ratio	0.557	0.358	0.000	0.209	0.628	0.902	1.000
Days-Active	27.388	29.526	1.000	5.000	15.000	40.000	120.000
Last-Period-Activity	0.299	0.314	0.000	0.000	0.235	0.403	1.000
Bonus-Cnt	0.024	0.203	0.000	0.000	0.000	0.000	3.000
Bonus-Amount	0.128	2.199	0.000	0.000	0.000	0.000	100.000
New-Player	0.047	0.211	0.000	0.000	0.000	0.000	1.000
Avg-Ticket-Size	6.349	14.792	0.000	0.971	2.049	5.212	200.000
Avg-Bet-Count	3.709	2.623	0.000	2.000	3.000	5.000	24.000
Churn	0.357	0.479	0.000	0.000	0.000	1.000	1.000
Distinct-Products	1.134	0.340	1.000	1.000	1.000	1.000	2.000
Balance	10.464	125.505	-21.200	0.000	0.000	0.000	4678.740
Ticket-Coverage	1.275	6.810	0.000	0.000	0.000	0.000	179.198

percentages of the l_1 norm, the performance measures may vary depending on the dataset used for training. For a typical default value of the regularization weight $\lambda = 1$, the performance measures' results do not vary across the percentage of l_1 norm used, but may vary significantly dependent on the datasets used for training the algorithm. However, the results of the proposed algorithm suggest that better performance may be achieved when increasing the value of the regularization weight λ , independent of the method used (i.e., the datasets used for training and testing the algorithm). More concretely, when increasing the penalty coefficient, there always exists an optimal combination of l_1 and l_2 norms in the elastic net penalty mixture for which the performance of the model is increased in comparison to the default mild regularization weight $\lambda = 1$, independent of the datasets used for training and testing.

Some key factors to assess the ability for correct classification made by the proposed algorithm included in Table 4, along with the feature importance, vary throughout the choice of the dataset used for training or testing in the simulation

Table 4 Elastic net performance measured by accuracy (ACC) and recall (REC) (in %) for $\lambda = 1$ with different training-test split (TTS) combinations

Design 1	I_1 / I_2											
	0.0/1.0	0.1/0.9	0.2/0.8	0.3/0.7	0.4/0.6	0.5/0.5	0.6/0.4	0.7/0.3	0.8/0.2	0.9/0.1	1.0/0.0	
TTS 90/10												
ACC	78.581	78.581	78.581	78.581	78.581	78.581	78.581	78.581	78.581	78.581	78.581	78.581
REC	64.130	64.130	64.130	64.130	64.130	64.130	64.130	64.130	64.130	64.130	64.130	64.130
TTS 80/20												
ACC	78.045	78.045	78.045	78.045	78.045	78.045	78.045	78.045	78.045	78.045	78.045	78.045
REC	62.799	62.799	62.799	62.799	62.799	62.799	62.799	62.799	62.799	62.799	62.799	62.799
TTS 70/30												
ACC	78.492	78.492	78.492	78.492	78.492	78.492	78.492	78.492	78.492	78.492	78.492	78.492
REC	63.469	63.469	63.469	63.469	63.469	63.469	63.469	63.469	63.469	63.469	63.469	63.469
TTS 60/40												
ACC	78.808	78.808	78.808	78.808	78.808	78.808	78.808	78.808	78.808	78.808	78.808	78.808
REC	65.363	65.363	65.363	65.363	65.363	65.363	65.363	65.363	65.363	65.363	65.363	65.363
TTS 50/50												
ACC	79.057	79.057	79.057	79.057	79.057	79.057	79.057	79.057	79.057	79.057	79.057	79.057
REC	66.642	66.642	66.642	66.642	66.642	66.642	66.642	66.642	66.642	66.642	66.642	66.642
TTS 40/60												
ACC	79.447	79.447	79.447	79.447	79.447	79.447	79.447	79.447	79.447	79.447	79.447	79.447
REC	65.667	65.667	65.667	65.667	65.667	65.667	65.667	65.667	65.667	65.667	65.667	65.667
TTS 30/70												
ACC	79.606	79.606	79.606	79.606	79.606	79.606	79.606	79.606	79.606	79.606	79.606	79.606
REC	63.890	63.890	63.890	63.890	63.890	63.890	63.890	63.890	63.890	63.890	63.890	63.890
TTS 20/80												
ACC	79.394	79.394	79.394	79.394	79.394	79.394	79.394	79.394	79.394	79.394	79.394	79.394
REC	61.993	61.993	61.993	61.993	61.993	61.993	61.993	61.993	61.993	61.993	61.993	61.993

study. The simulation study considers four different combinations of different training and testing data choices (called methods) based on the original dataset and bootstrapped samples (with replacements) from the original dataset. To resemble the features of the original dataset, the bootstrapped sample keeps the same percentage of churners as the original data set. However, we also consider the bootstrapped sample with the same percentage of the two classes for both the training and the testing dataset to compare the algorithm's performance with a perfectly balanced dataset. This set-up is known to perform best in classification problems using logistic regression models with a predefined classification threshold (usually, this threshold is set to 0.5 in common logistic regression implementations).

In addition to accuracy, we explore the algorithm's predictive ability through an additional performance measure recall for all combinations of the datasets used for training and testing in Table 4. According to the simulation study, such combination of datasets for training versus testing seems to have the potential to favor either the no-regularization case or the proposed algorithm depending on the performance measure used. Already in the case of a standard, moderate penalty $\lambda = 1$, we find an increase in classification accuracy of the proposed algorithm for all methods independent of the percentage combination of norms l_1 and l_2 used. In terms of the recall performance measure, an increase of the classification ability is much stronger in comparison to the accuracy performance, and the magnitude of that increase depends on the method used. More specifically, when the original dataset is used for training and the bootstrapped sample (with the same percentage of churners as in the original or with a balanced percentage), the proposed algorithm produced an increase of 9% in classification ability relative to the no-regularization case for a specific mixture penalization.

Compared to the mild regularization case of $\lambda = 1$, the proposed algorithm finds a mixture of l_1 and l_2 norms, which may increase the classification ability by 3.6% for the same dataset used. Moreover, when the bootstrapped sample (with the same percentage of churners as in the original or with a balanced percentage) is used for training and the original data set for testing, the proposed algorithm produces an increase of more than 10% in classification ability relative to the no-regularization case for a specific mixture penalization. However, when compared to the mild regularization case of $\lambda = 1$, the proposed algorithm finds a mixture of l_1 and l_2 norms, which may increase the classification ability by more than 13% for the same dataset used. In terms of feature importance, the proposed algorithm puts in the first place some features that were completely neglected in the no-regularization case and completely rearranges the weights of their importance.

Figures 1 and 2 show that the proposed algorithm can detect at least one feature that the no-regularization case completely ignores. The opposite is not true. Namely, the features that do not appear significant in the proposed algorithm are also of minor importance in the no-regularization case. Depending on the performance measure used for evaluating the algorithm performance, the results may also significantly differ in terms of feature importance.

Results considering the accuracy performance measure are shown in Tables 10 and 11. First, cross-validation shows that in the case of the elastic net regularization, the best accuracy is achieved when training the algorithm on the highest percentage

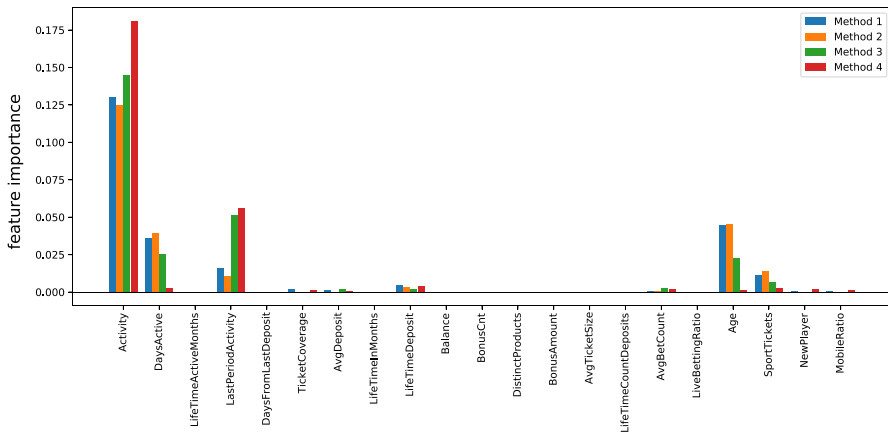


Fig. 1 Feature importance based on accuracy for an optimal mixture of penalized LR models

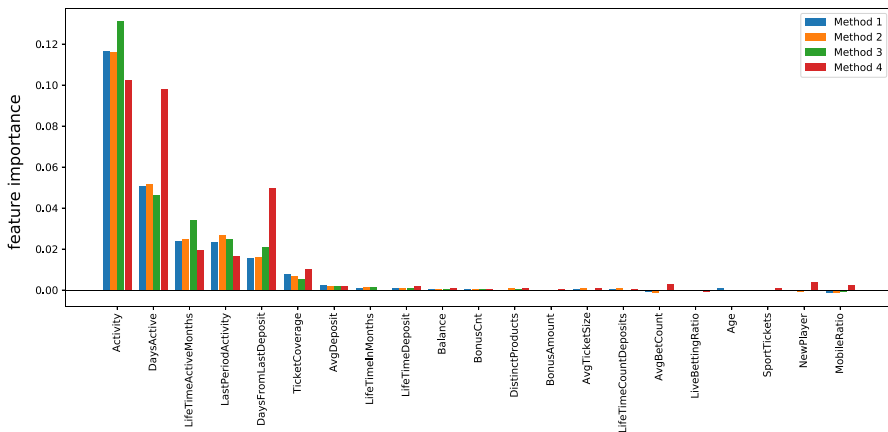


Fig. 2 Feature importance based on accuracy for the conventional LR model without penalty

split of the original dataset with a significant increase of the regularization weight. On the other hand, the elastic net penalty's mixing coefficient is of secondary importance for non-optimal sparse solutions. Namely, if the penalty weight is not set optimally, the accuracy remains unaltered by changing the percentage of the l_1 norm in the elastic net (Table 4). Compared to the no regularization case, the proposed algorithm can increase the accuracy score by 2%.

Second, the proposed algorithm demonstrates that when the highest accuracy is the ultimate goal, the numbers of days from the last deposit date with respect to the date of measurement (*Days-From-Last-Deposit*) has the most pronounced effect on the probability of churning from the company. Moreover, the proposed algorithm reveals the importance of another feature in reducing customer churn that the no regularization case completely neglects. Namely, the second most important feature according to our study is the number of sports tickets placed in the observed period

(*Sport-Tickets*), and not surprisingly the activity measure counting the number of active months in customer lifetime that puts more weight on recent history (*Activity*). This finding suggests that just being a present customer in the most recent period is not the most important behavior that the company expects in order to retain its customers. On the contrary, loyalty and fidelity kind of behavior are emphasized even more clearly with these findings. The company would expect its customers to diversify their deposits over a lifetime and not be present just over some specific period in time and almost completely absent otherwise.

The robustness of the results above is additionally confirmed in the simulation study in Sect. 3.3: varying the datasets used for training versus validating the algorithm. When considering the no-regularization case, we find that both the number of active months in the customer's lifetime, putting more weight on recent history (*Activity*), and the number of days when a customer placed at least one bet in the observed period (*Days-Active*) have the most pronounced effects on retaining customers, independently of the dataset used for training and validation. Consequently, even if we explore the possibility of using enlarged datasets for training and validation, the no-regularization case model is not capable of detecting the number of sports tickets placed in the observed period (*Sport-Tickets*) as a potentially influential feature when targeting the highest accuracy. Moreover, the moderate importance of the number of months in which a customer has placed at least one bet (*Life-Time-Active-Months*) on the probability of churning from the company would potentially suggest that a customer does not have to be active every single month and still will not necessarily churn away from the company.

Some of these findings are additionally confirmed when considering maximizing the recall performance measure, as shown in Tables 10 and 11. First, through a cross-validation analysis in the mixture penalized case, maximal recall is achieved when training the algorithm on high split percentages of the original dataset and significantly increasing the regularization weight. Furthermore, if the regularization weight is not optimally set when considering sparse solutions, then increasing the percentage of the l_1 norm in the elastic net does not influence the recall performance (Table 4). Compared with the no-regularization case, the proposed algorithm can increase the recall performance by more than 5 percentage points.

Second, the proposed algorithm demonstrates that when targeting the highest recall, the numbers of days from the last deposit date with respect to the date of measurement (*Days-From-Last-Deposit*) and the number of sports tickets placed in the observed period (*Sport-Tickets*) become the most influential features, followed by the number of customer transfers to the company's account (*Life-Time-Count-Deposits*). Moreover, the influence of these features is even more pronounced when considering the recall performance measure with respect to accuracy. Additionally, even though all possibilities of using enlarged datasets for training and validation are explored, the no regularization model cannot detect the number of sports tickets placed in the observed period as a potentially influential feature in the case of targeting the highest recall.

3.3 Validation and Robustness

The simulation study covers four different methods, each differing with respect to the dataset used for training versus testing. We do not assume any specific distribution of the data, but we instead choose to sample from the originally observed data, aiming to work with data that have happened in the business life of the company. We bootstrap from the original sample (with replacements) a sample length ten times bigger in two ways. One bootstrapped sample has to have the same percentage of churners as in the original dataset in order to have potential to resemble similar features of the observed data set. The second bootstrapped sample is chosen in such a way as to get a balanced dataset (50% of each of the two categories). The reason for the latter is twofold. First, typical logistic regression algorithms work with a default threshold of 0.5, which is known to perform the best in the case of a balanced dataset. Second, depending on the percentage of the specific binary category in a dataset, in terms of accuracy, logistic regression algorithms are empirically known to perform best when setting a threshold equal to that percentage. More specifically, we consider the following four methods.

- (1) Method 1 uses the original dataset of length n_1 for training and the bootstrapped sample (with replacements) of size $10n_1$ with the same percentage of churners as in the original dataset for testing.
- (2) Method 2 uses the original dataset of length n_1 for training and the bootstrapped sample (with replacement) of size $10(n_1 + 1)$ with the balanced percentage of churners for testing.
- (3) Method 3 uses the bootstrapped sample (with replacements) of size $10n_1$ with the same percentage of churners as in the original dataset for training and the original dataset of length n_1 for testing.
- (4) Method 4 uses the bootstrapped sample (with replacements) of size $10(n_1 + 1)$ with the balanced percentage of churners for training and the original dataset of length n_1 for testing.

The algorithm performance for such simulated designs is analyzed by varying the following parameters: the performance measure, the percentage of churners, the percentage of the l_1 norm in the elastic net specification, and the regularization weight through a cross-validation setting. In addition to the accuracy performance measure as a standard, we analyze the recall performance measure in terms of choosing the optimal penalty weight λ . In the business context of churning classification, we prefer a classifier that does not reject many true churners in comparison to a classifier that rejects many true churners—namely, to have a high recall. The percentage of churners considered is the same percentage as in the original dataset (35.74%) and 50%, with the latter corresponding to the case of a balanced dataset. In other words, for each instance the algorithm computes a score based on a decision function. If the estimated probability that an instance belongs to a particular class is greater than 50%, then the model predicts that the instance belongs to that class (churners). According to this threshold of 0.5, all instances beyond the

decision boundary have an over 50% chance of being churners according to the model.

There are two commonly-used techniques for handling imbalanced data in churn prediction: (1) resampling methods that balance the data before training, and (2) cost-sensitive learning methods that adjust the relative costs of the errors during training (see (Nguyen & Duong, 2021), for example). The proposed method in this study shows that the penalty parameter λ is of significantly higher importance than the regularization parameter α when dealing with imbalanced data for churn prediction. In addition, we apply the resampling method as listed above, and the results confirm that both predictive performance and feature importance change substantially when tuning the penalty level in comparison to the amount of an additional non-necessarily differentiable regularization function added to the loss-function. In this empirical study, we also apply cross-validation (i.e., resampling) with different training-test split (TTS) combinations of the data to show the robustness of the proposed method (Tables 5, 6, 7).

3.4 Validation of Results

The validation of results appears given in Tables 8, 9, 10, 11. The proposed algorithm becomes preferable to the simple no-regularization logistic regression, because the regularization weight increases independently of the method used. From Table 8 we see that all four methods suggest that both accuracy and recall performance measures do not change when the elastic net varying parameter representing the percentage of the l_1 norm changes, as long as some regularization weight is used. The same conclusion holds when validating the algorithm on the original dataset that is split dependent on the percentage of the original dataset used for training. Our empirical analyses jointly with the proposed algorithm show that independent of the method used, the performance measure changes by varying the α parameter in the elastic net specification for optimal weights λ^* chosen based on the proposed algorithm. Thus, the corresponding performance measures achieve their maximal value for a set of quite high values of the regularization weight, and then the performance measure falls even when increasing the regularization weight λ . The simulation study reveals that the accuracy performance measure may be increased when using the proposed algorithm instead of the no-regularization model by more than 2% based on the

Table 5 Performance of logistic regression without penalty measured by accuracy and recall (in %)

Design	Accuracy	Recall
TTS 90/10	78.715	63.043
TTS 80/20	78.313	61.510
TTS 70/30	79.161	62.362
TTS 60/40	79.176	62.291
TTS 50/50	79.405	62.472
TTS 40/60	79.692	62.485
TTS 30/70	80.218	63.152
TTS 20/80	80.566	63.996

Table 6 Performance of optimal regularization weights λ^* (10^4 units) measured by accuracy (ACC) and recall (REC)

Cross-validation	l_1 / l_2											
	0.0/1.0	0.1/0.9	0.2/0.8	0.3/0.7	0.4/0.6	0.5/0.5	0.6/0.4	0.7/0.3	0.8/0.2	0.9/0.1	1.0/0.0	
TTS 90/10												
	ACC	35.938	2.783	2.783	1.000	1.000	1.000	1.000	1.000	1.000	0.215	
REC	10.000	2.783	2.783	2.783	1.000	1.000	1.000	1.000	1.000	1.000	1.000	
TTS 80/20												
	ACC	2.783	0.010	0.010	0.100	0.001	0.001	0.001	0.001	0.001	0.001	
REC	1.000	2.783	2.783	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	
TTS 70/30												
	ACC	1.000	2.783	1.000	0.100	1.000	0.215	0.215	0.100	0.215	0.215	
REC	1.000	2.783	1.000	1.000	1.000	1.000	1.000	0.100	0.100	0.215	0.215	
TTS 60/40												
	ACC	2.783	0.215	1.000	0.001	0.001	0.100	0.100	0.100	0.010	0.215	
REC	1.000	2.783	1.000	1.000	1.000	1.000	0.215	0.100	0.100	0.100	0.215	
TTS 50/50												
	ACC	2.783	1.000	0.100	0.010	0.100	0.100	0.100	0.100	0.100	0.100	
REC	0.100	2.783	0.100	0.100	1.000	1.000	1.000	0.215	0.215	0.100	0.215	
TTS 40/60												
	ACC	10.000	1.000	0.215	0.215	0.100	0.100	0.100	0.100	0.100	0.100	
REC	0.100	1.000	1.000	1.000	0.215	0.100	0.215	0.100	0.100	0.215	0.215	
TTS 30/70												
	ACC	1.000	1.000	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	
REC	–	1.000	1.000	1.000	0.100	0.215	0.100	0.100	0.100	0.100	0.100	
TTS 20/80												
	ACC	–	–	–	–	0.100	0.100	0.100	0.100	0.100	0.010	
REC	–	1.000	0.215	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	

Table 7 Relative changes (in %) in accuracy (ACC) and recall (REC) of the proposed algorithm with a regularization $\lambda = 1$

Cross-validation	I_1 / I_2										
	0.0/1.0	0.1/0.9	0.2/0.8	0.3/0.7	0.4/0.6	0.5/0.5	0.6/0.4	0.7/0.3	0.8/0.2	0.9/0.1	1.0/0.0
TTS 90/10											
	ACC	2.214	0.851	0.681	0.681	0.851	0.681	0.681	0.681	0.512	0.681
	REC	0.000	4.521	6.215	5.651	5.651	6.215	6.215	5.651	5.085	4.521
TTS 80/20											
	ACC	0.086	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
	REC	6.736	2.933	4.107	3.186	3.812	4.107	4.107	3.519	2.640	2.054
TTS 70/30											
	ACC	0.056	0.397	0.511	0.340	0.227	0.340	0.397	0.171	0.626	0.511
	REC	0.000	4.457	3.682	4.651	4.845	3.295	3.101	2.132	4.070	3.876
TTS 60/40											
	ACC	0.085	0.085	0.213	0.043	0.000	0.085	0.000	0.000	0.000	0.085
	REC	0.000	2.706	3.135	2.706	2.564	2.137	1.140	1.282	1.424	3.277
TTS 50/50											
	ACC	0.577	0.034	0.034	0.102	0.034	0.102	0.068	0.068	0.034	0.034
	REC	0.000	2.234	2.123	2.234	2.234	1.116	2.011	2.125	1.229	2.123
TTS 40/60											
	ACC	0.028	0.167	0.196	0.140	0.281	0.281	0.308	0.252	0.252	0.140
	REC	0.000	2.516	2.982	2.423	2.330	1.864	2.237	2.237	3.076	2.889
TTS 30/70											
	ACC	0.024	0.192	0.072	0.350	0.337	0.121	0.192	0.192	0.240	0.168
	REC	0.000	2.806	2.229	1.321	1.485	1.982	2.310	2.310	2.640	2.806
TTS 20/80											
	ACC	0.000	0.000	0.000	0.000	0.000	0.084	0.127	0.063	0.021	0.000
	REC	0.000	0.752	1.203	0.752	1.203	1.729	1.879	1.653	1.428	1.128

Table 8 Elastic net performance measured by accuracy (ACC) and recall (REC) (in %) with $\lambda = 1$ for the augmented data

	l_1 / l_2										
	0.0/1.0	0.1/0.9	0.2/0.8	0.3/0.7	0.4/0.6	0.5/0.5	0.6/0.4	0.7/0.3	0.8/0.2	0.9/0.1	1.0/0.0
Method 1											
ACC	80.471	80.471	80.471	80.471	80.471	80.471	80.471	80.471	80.471	80.471	80.471
REC	65.777	65.777	65.777	65.777	65.777	65.777	65.777	65.777	65.777	65.777	65.777
Method 2											
ACC	77.100	77.100	77.100	77.100	77.100	77.100	77.100	77.100	77.100	77.100	77.100
REC	65.447	65.447	65.447	65.447	65.447	65.447	65.447	65.447	65.447	65.447	65.447
Method 3											
ACC	79.577	79.577	79.577	79.577	79.577	79.577	79.577	79.577	79.577	79.577	79.577
REC	60.510	60.510	60.510	60.510	60.510	60.510	60.510	60.510	60.510	60.510	60.510
Method 4											
ACC	80.541	80.541	80.541	80.541	80.541	80.541	80.541	80.541	80.541	80.541	80.541
REC	76.733	76.733	76.733	76.733	76.733	76.733	76.733	76.733	76.733	76.733	76.733

Table 9 Performance of logistic regression without penalty measured by accuracy and recall (in %) for the augmented data

	Accuracy	Recall
Method 1	79.646	62.495
Method 2	76.044	62.608
Method 3	79.711	62.458
Method 4	79.938	78.531

method used in the study. Furthermore, both accuracy and recall performance of the proposed algorithm rise even with respect to the regularization weight case under the elastic net specification for a standard default case of the penalty coefficient being 1.

The results in terms of optimal penalty coefficients are shown in Table 10. We link the optimal weight λ^* from the proposed algorithm to the varying parameter α from the elastic net specification (9). Independent of the method used, the proposed algorithm is capable of finding an optimal combination of the regularization weight λ^* , and the varying parameter α for which the performance measures, both in terms of accuracy and recall, may be increased in comparison to the no-regularization case. The optimal regularization weight is eventually not equal for all the percentages of norm l_1 used in the elastic net penalty. For a different value of the elastic net penalty and in every study method used, the algorithm can recognize an optimal regularization weight λ^* that achieves the same level of a performance measure in question up to the fourth decimal.

Table 11 presents how the proposed algorithm can improve accuracy and recall in comparison to the regularization case with the default penalty value $\lambda = 1$. The simulation study shows that specific combinations of the regularization weight λ and the percentage of l_1 norm α in the elastic net penalty have been found to outperform elaborated weighted combinations in repeated empirical applications.

In terms of feature importance, Figs. 1 and 3 illustrate that the proposed algorithm reorders their weights and their order of importance with respect to the penalized cases and Figs. 2 and 4 for the non-penalized cases based on accuracy and recall, respectively. In this work we find a way to calculate the feature importance for regularized logistic regression, which is not given by default in classical machine learning algorithms as discussed in Sect. 2.2. Figure 1 demonstrates that independent of the method used and when the highest accuracy is the ultimate goal, the numbers of days from the last deposit date with respect to the date of measurement (*Days-From-Last-Deposit*) has the most pronounced effect on the probability of churning from the company. Furthermore, the proposed algorithm reveals the importance of another feature in reducing customer churn that the no-regularization case completely neglects; that is, when considering Method 1 and Method 2 (the algorithm is trained on the original dataset) in our study, the second most important feature is the number of sports tickets placed in the observed period (*Sport-Tickets*). However, when considering the bootstrapped samples for training in our study (Method 3 and Method 4), the number of days when a customer placed at least one bet (*Days-Active*) is found to be the second most dominant feature. This finding directly reflects the results of the no regularization case in which the same feature is found to be the

Table 10 Performance of the optimal regularization weights λ^* (10^4 units) measured by accuracy (ACC) and recall (REC) for the augmented data

l_1 / l_2		0.0/1.0	0.1/0.9	0.2/0.8	0.3/0.7	0.4/0.6	0.5/0.5	0.6/0.4	0.7/0.3	0.8/0.2	0.9/0.1	1.0/0.0
Method 1												
ACC	1.000	0.100	0.010	0.010	0.010	0.010	0.010	0.010	0.100	0.010	0.010	0.010
REC	–	2.783	2.783	2.783	2.783	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Method 2												
ACC	1.000	2.783	2.783	1.000	1.000	1.000	1.000	1.000	0.100	0.215	0.215	0.215
REC	–	2.783	2.783	2.783	2.783	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Method 3												
ACC	1000.000	100.000	35.938	35.938	10.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000
REC	1000.000	100.000	35.938	35.938	35.938	35.938	35.938	10.000	10.000	10.000	10.000	10.000
Method 4												
ACC	10.000	2.783	1.000	1.000	1.000	1.000	1.000	0.215	0.215	0.215	0.215	0.215
REC	1000.000	100.000	63.096	46.416	46.416	46.416	46.416	19.307	19.307	19.307	19.307	19.307

Table 11 Relative changes (in %) in accuracy (ACC) and recall (REC) of the proposed algorithm with a regularization $\lambda = 1$ for the augmented data

	l_1 / l_2										
	0.0/1.0	0.1/0.9	0.2/0.8	0.3/0.7	0.4/0.6	0.5/0.5	0.6/0.4	0.7/0.3	0.8/0.2	0.9/0.1	1.0/0.0
Method 1											
ACC	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.020	0.000	0.000	0.000
REC	0.000	2.171	3.554	2.729	3.237	3.456	3.446	3.282	2.758	2.695	2.347
Method 2											
ACC	0.000	0.304	0.278	0.413	0.473	0.355	0.208	0.172	0.170	0.157	0.285
REC	0.000	2.009	3.720	2.884	3.204	3.572	3.621	3.367	2.913	2.799	2.528
Method 3											
ACC	0.185	0.606	0.925	0.589	0.538	1.061	1.061	0.909	0.841	0.808	0.808
REC	1.858	13.312	12.879	13.251	12.940	10.897	10.464	12.198	13.436	13.436	13.621
Method 4											
ACC	0.016	0.100	0.100	0.067	0.133	0.067	0.067	0.050	0.067	0.099	0.133
REC	4.590	7.520	8.887	8.935	8.398	7.715	8.154	9.082	9.082	8.887	8.448

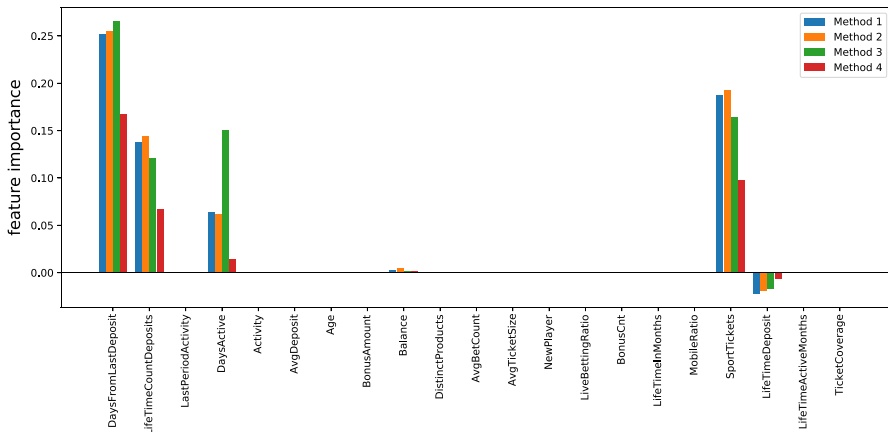


Fig. 3 Feature importance based on recall for an optimal mixture of penalized LR models

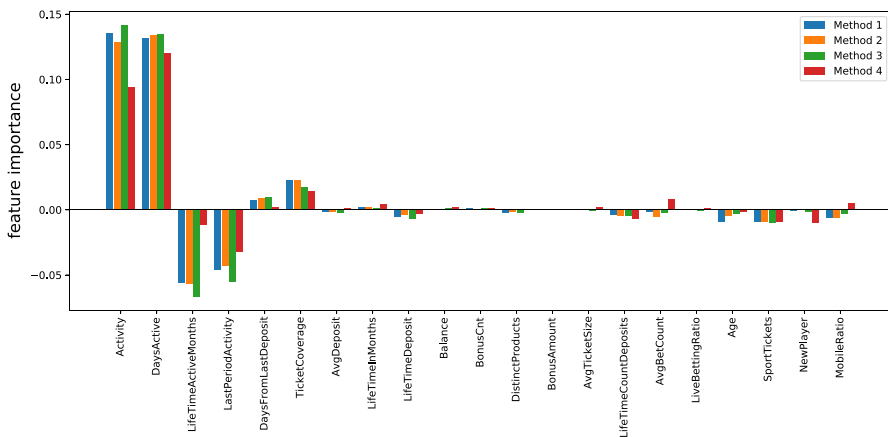


Fig. 4 Feature importance based on recall for the conventional LR model without penalty

second most dominant feature, but of much higher importance over all four simulation methods considered.

The proposed algorithm is thus not only capable of effectively improving the accuracy of the classification, but also able to shed more light on the feature of the data. Depending on the enlarged dataset used for training the algorithm, the proposed mixture penalty method finds all the most important features established in the no-regularization case and, more importantly, discovers at least one new influential feature completely neglected in the no-penalty case. This result's robustness is shown when considering the overall recall measure for all four methods. Figure 3 shows that the two most important features discovered by the proposed algorithm (*Days-From-Last-Deposit* and *Sport-Tickets* respectively) are found to be the most influential features independently of the method used in the

simulation study, with even more pronounced importance in comparison to performance accuracy.

3.5 Discussion

Our major contribution to the literature is to introduce a novel machine learning-based algorithm with sparse regularization for explainable customer churn modelling to support operational decision-making. We evaluate the predictive performance of the proposed sparse regularization method to model recency, frequency, and monetary value variables. When comparing methods for churn prediction with different datasets, the main challenge is caused by the specific case considered in order to understand the company's plan to retain its clients, while at the same time taking into consideration that acquiring new customers is much more expensive than retaining customers who might leave. To the best of our knowledge, the only research comparable to our goals aiming to reconcile efficient predictive performance and explainable machine learning algorithms for churn prediction is De Bock and De Caigny (2021) that consider the sparse group lasso regularization to enhance interpretability through structured regularization.

We emphasize even more deeply feature engineering by measuring feature performance with both positive and negative values; that is, in our analysis we look at how much the score (i.e., accuracy and precision) decreases when a corresponding feature is not available. Consequently, negative values indicate that the score increases when permuting the feature since feature permutation actually improves the score. For both managerial applicability and machine learning explainability, deep learning methods may not tackle this issue. We still need to pay attention to the positive sign and magnitude of feature importance. In this way we may evaluate how important the feature is not only based on the performance scoring, but also based on how each feature can individually contribute to the decision making process.

We consider the features in this work based on recency, frequency, and monetary value variables (see Zhang et al., 2015) and significantly enable the flexibility of the proposed regularized logistic regression classifier that has been commonly used with demographic variables when dealing with high dimensional features. We measure the recency of the interaction between the company and its clients as well as frequency of these interactions and additionally incorporate the specific nature of such interactions, since the customers do not need to be active frequently yet still stay. To highlight these interactions we introduce three different features: Life-Time-In-Months, Life-Time-Active-Months, and Activity. In addition, we explain how to use time-dependent features (such as Life-Time-In/Active-Months, Days-From-Last-Deposit, and Life-Time based factors) in classification problems when extracting time-varying information as new features. However, few studies in the literature incorporate both static and dynamic variables for classification algorithms by characterizing the client activities and the interaction between the client and the company services at same time as what we have conducted in this study, which enhances the interpretability of algorithms.

4 Conclusion

Focusing on how to effectively distinguish consumers with relatively high loyalty, this research proposes an optimized logistics classification method that considers both l_1 and l_2 norm penalization to reduce overfitting of unbalanced data. In the method we proposed, we fully consider the impact of unbalanced data on the algorithm. In other words, when the target customers account for a relatively small proportion of the overall sample, some models will produce overfitting.

Although the l_1 and l_2 norm penalties can be used separately to adapt to different data features, it is reasonable to mix these two penalties when the features cannot be clearly perceived. Our proposed machine learning-based inference approach corrects for overfitting with unbalanced data and is less prone to noise. It may also tackle a large number of possibly highly correlated explanatory variables controlling for multicollinearity. We provide the expected performance gain compared to no-regularization linear benchmark models and manage to deliver better performance in terms of accuracy and recall measures. For such an optimal penalty value and the corresponding mixing combination of norms, we identify the most important features that maximize specified performance.

We consider the optimization in two ways. First, for any given penalty value in the loss function, when combined with the chosen mixing coefficient of l_1 and l_2 norms, we minimize the binary cross-entropy error. Second, among all the combinations of the penalty value and mixing norms, coefficient minimizing the targeted error, we choose the optimal one maximizing the performance measure of interest. Moreover, we consider the standard specification of the used performance measure and capture the impact of mixture terms on optimal decision-making.

We conducted an empirical study with the proposed method. Our empirical results confirm the method's effectiveness and show that the mixture penalty method indeed improves classification accuracy. However, it is necessary to carefully consider the different effects of the two types of errors and the various costs incurred subsequently.

Identifying consumer loyalty is theoretically more complex, because consumer preferences and behavior are affected by many factors and may change over time. Identifying these factors requires a more extensive dataset and related analysis techniques. Our proposed method could be extended for further analysis in at least two ways. First, modeling abnormal behavior observed from the data enables us to evaluate the function of customer relationship management (CRM) even more. Second, using machine-learning methods without considering the probability model could be a useful attempt to reduce dimensionality of big data and provide a foundation for more interpretable models. It is thus attractive for private institutions to enrich their marketing tools by not only widening the means of promotion, but also effectively controlling their costs.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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